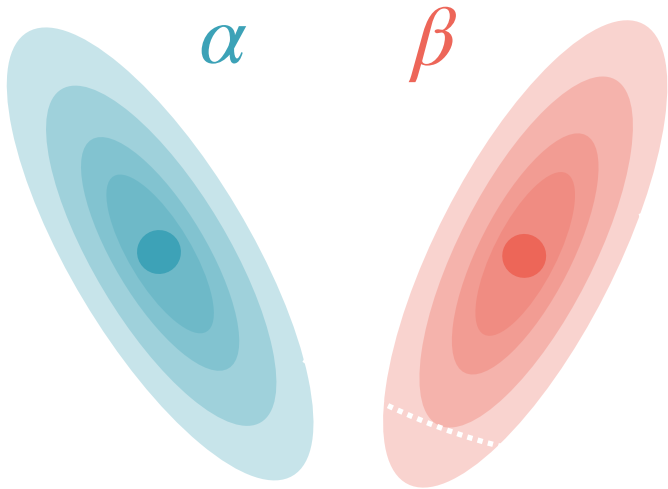
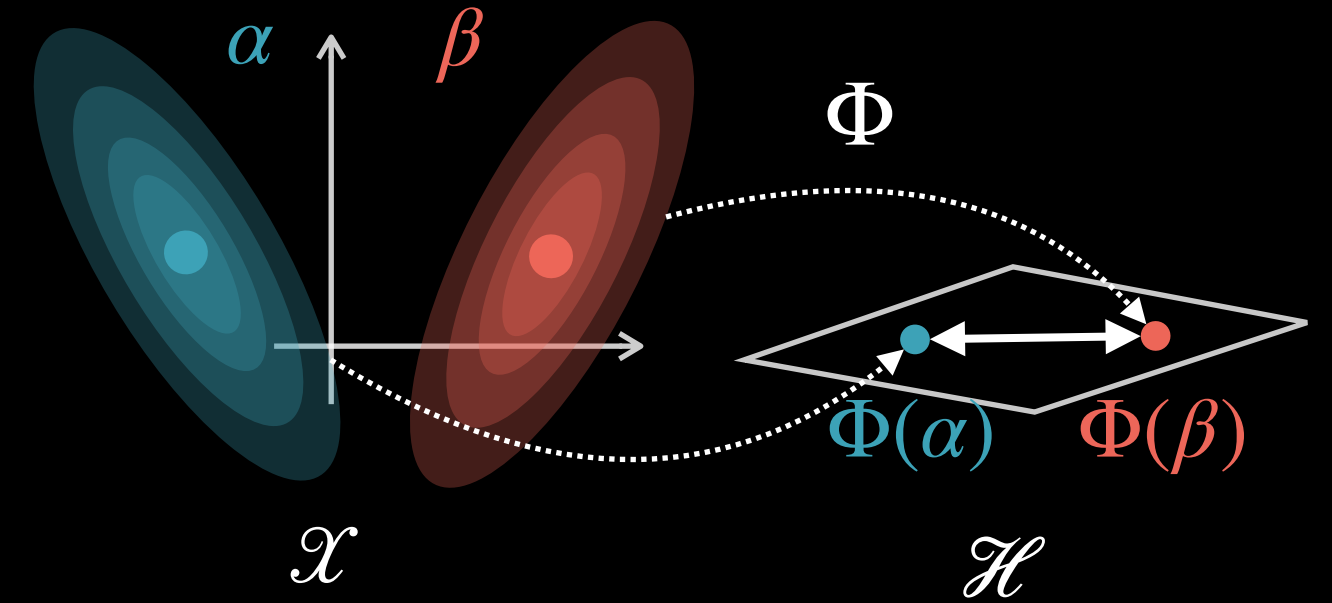




Hilbertian Metrics

- **Definition:** A metric d on a set $\mathcal{X}$ is Hilbertian if there exists a Hilbert space $\mathcal{H}$ and an embedding $\Phi : \mathcal{X} \rightarrow \mathcal{H}$ such that $d(x, y) = \|\Phi(x) - \Phi(y)\|_{\mathcal{H}}$
- **Intuition:** ~ the geometry of d is just Euclidean “hidden” in a possibly (∞ -dim) feature space.
- **Importance:**
 - Hilbertian metrics admit positive definite kernels, i.e., can write them as $d(x, y)^2 = k(x, x) + k(y, y) - 2k(x, y)$ for a PSD kernel k , eg by taking $k(x, y) = \langle \Phi(x), \Phi(y) \rangle$
 - Many efficient algorithms (kernel methods, random projections, convex opt) rely on this.
- **Examples:**
 - Hilbertian: MMD (by construction!), Energy, Hellinger, χ^2
 - Non-Hilbertian: W_p in $d \geq 2$, TV, KL, JS (and most other f -divergences)
- **Takeaway:** Hilbertian $\Rightarrow$ kernelizable $\Rightarrow$ statistically efficient; non-Hilbertian $\Rightarrow$ harder to approximate



When to use what?